

MongoDB-Student Collection

-Show dbs

```
> show dbs
< admin      40.00 KiB
  config     36.00 KiB
  local      40.00 KiB
  studentDB   8.00 KiB
```

-use StudentDB

```
show actions studentDB
< switched to db studentDB
> db
< studentDB
> show collections
< stuedents
> db.createCollection("students")
< { ok: 1 }
> db.students.drop()
< true
```

insertMany

```

> db.students.insertMany([
  { rollNo: 101, name: "John", age: 20, department: "CSE", marks: 85, city: "Hyderabad" },
  { rollNo: 102, name: "Alice", age: 21, department: "CSE", marks: 90, city: "Vijayawada" },
  { rollNo: 103, name: "Rahul", age: 20, department: "ECE", marks: 75, city: "Hyderabad" },
  { rollNo: 104, name: "Priya", age: 22, department: "IT", marks: 88, city: "Warangal" },
  { rollNo: 105, name: "Kiran", age: 21, department: "CSE", marks: 95, city: "Hyderabad" },
  { rollNo: 106, name: "Anu", age: 20, department: "IT", marks: 82, city: "Guntur" },
  { rollNo: 107, name: "Ravi", age: 23, department: "ECE", marks: 68, city: "Nellore" },
  { rollNo: 108, name: "Sita", age: 21, department: "CSE", marks: 91, city: "Hyderabad" },
  { rollNo: 109, name: "Arun", age: 22, department: "AI&DS", marks: 87, city: "Kurnool" },
  { rollNo: 110, name: "Meena", age: 20, department: "AI&DS", marks: 93, city: "Hyderabad" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8c015a8e356ef8a07b875b'),
    '1': ObjectId('6a8c015a8e356ef8a07b875c'),
    '2': ObjectId('6a8c015a8e356ef8a07b875d'),
    '3': ObjectId('6a8c015a8e356ef8a07b875e'),
    '4': ObjectId('6a8c015a8e356ef8a07b875f'),
    '5': ObjectId('6a8c015a8e356ef8a07b8760'),
    '6': ObjectId('6a8c015a8e356ef8a07b8761'),
    '7': ObjectId('6a8c015a8e356ef8a07b8762'),
    '8': ObjectId('6a8c015a8e356ef8a07b8763'),
    '9': ObjectId('6a8c015a8e356ef8a07b8764')
  }
}

```

-InsertOne

```

> MONGOSH
> db.students.insertOne({})
< {
  acknowledged: true,
  insertedId: ObjectId('6a8c01d38e356ef8a07b8765')
}
> db.students.insertOne({
  rollNo: 111,
  name: "Vijay",
  age: 21,
  department: "CSE",
  marks: 86,
  city: "Hyderabad"
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a8c01e18e356ef8a07b8766')
}

```

-InsertMany

>_MONGOSH

```
db.students.insertMany([
  { rollNo: 112, name: "Pooja", age: 20, department: "IT", marks: 84 },
  { rollNo: 113, name: "Ramesh", age: 22, department: "ECE", marks: 79 }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8c01ec8e356ef8a07b8767'),
    '1': ObjectId('6a8c01ec8e356ef8a07b8768')
  }
}
```

-Find

>_MONGOSH

> db.students.find()

```
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875d'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
```

>_MONGOSH

```
{
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8760'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8761'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8762'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
```

>_MONGOSH

```
    rollNo: 110,  
    name: 'Meena',  
    age: 20,  
    department: 'AI&DS',  
    marks: 93,  
    city: 'Hyderabad'  
  }  
  {  
    _id: ObjectId('6a8c01d38e356ef8a07b8765')  
  }  
  {  
    _id: ObjectId('6a8c01e18e356ef8a07b8766'),  
    rollNo: 111,  
    name: 'Vijay',  
    age: 21,  
    department: 'CSE',  
    marks: 86,  
    city: 'Hyderabad'  
  }  
  {  
    _id: ObjectId('6a8c01ec8e356ef8a07b8767'),  
    rollNo: 112,  
    name: 'Pooja',  
    age: 20,  
    department: 'IT',  
    marks: 84  
  }  
  {  
    _id: ObjectId('6a8c01ec8e356ef8a07b8768'),  
    rollNo: 113,  
    name: 'Ramesh',  
    age: 22,  
    department: 'ECE',  
    marks: 79  
  }  
}
```

> _MONGOSH

> db.students.find({ department: "CSE" })

< {

```
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
```

}

{

```
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
```

}

{

```
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
```

}

{

```
  _id: ObjectId('6a8c015a8e356ef8a07b8762'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
```



>_MONGOSH



```
> db.students.find({ name: "John" })
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
> db.students.countDocuments()
< 14
> db.students.countDocuments({ department: "CSE" })
< 5
> db.students.find({ marks: { $gt: 80 } })
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
```

>_MONGOSH

```
_id: ObjectId('6a8c015a8e356ef8a07b8760'),
rollNo: 106,
name: 'Anu',
age: 20,
department: 'IT',
marks: 82,
city: 'Guntur'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8762'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8763'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8764'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
```

>_MONGOSH

```
> db.students.find({ marks: { $gte: 80 } })
```

```
< {
```

```
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
```

```
}
```

```
{
```

```
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
```

```
}
```

```
{
```

```
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
```

```
}
```

```
{
```

```
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
```

>_MONGOSH

```
> db.students.find({ marks: { $lt: 80 } })
```

```
< {
```

```
  _id: ObjectId('6a8c015a8e356ef8a07b875d'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
```

```
}
```

```
{
```

```
  _id: ObjectId('6a8c015a8e356ef8a07b8761'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
```

```
}
```

```
{
```

```
  _id: ObjectId('6a8c01ec8e356ef8a07b8768'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
```

```
}
```

```
>_MONGOSH
```

```
> db.students.find({ marks: { $lte: 80 } })
```

```
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875d'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8761'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c01ec8e356ef8a07b8768'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
```

```
> db.students.find({ department: { $ne: "CSE" } })
```

```
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875d'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',

```

>_MONGOSH



```
{
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8760'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8761'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8763'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
```

>_MONGOSH

```
}  
> db.students.find({ marks: { $eq: 90 } })  
< {  
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),  
  rollNo: 102,  
  name: 'Alice',  
  age: 21,  
  department: 'CSE',  
  marks: 90,  
  city: 'Vijayawada'  
}  
> db.students.find({  
  $and: [  
    { department: "CSE" },  
    { marks: { $gt: 85 } }  
  ]  
})  
< {  
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),  
  rollNo: 102,  
  name: 'Alice',  
  age: 21,  
  department: 'CSE',  
  marks: 90,  
  city: 'Vijayawada'  
}  
{  
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),  
  rollNo: 105,  
  name: 'Kiran',  
  age: 21,  
  department: 'CSE',  
  marks: 95,  
  city: 'Hyderabad'  
}
```

>_MONGOSH

```
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8762'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c01e18e356ef8a07b8766'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
  marks: 86,
  city: 'Hyderabad'
}
> db.students.find({
  $or: [
    { department: "CSE" },
    { department: "IT" }
  ]
})
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
```

>_MONGOSH

```
}
> db.studennts.find({})
<
> db.stuudennt.find({marks:{$not:{$gt:80 }}})
<
> db.students.find({
  marks: { $not: { $gt: 80 } }
})
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875d'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8761'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c01d38e356ef8a07b8765')
}
{
  _id: ObjectId('6a8c01ec8e356ef8a07b8768'),
  rollNo: 113,
  name: 'Ramesh',
  age: 22,
  department: 'ECE',
  marks: 79
}
```

>_MONGOSH

```
> db.students.find({
  $nor: [
    { department: "CSE" },
    { marks: { $gt: 90 } }
  ]
})
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875d'),
  rollNo: 103,
  name: 'Rahul',
  age: 20,
  department: 'ECE',
  marks: 75,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8760'),
  rollNo: 106,
  name: 'Anu',
  age: 20,
  department: 'IT',
  marks: 82,
  city: 'Guntur'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8761'),
```

```
> _MONGOSH
```

```
> db.students.find({
  department: { $in: ["CSE", "IT"] }
})

< {
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 85,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),
  rollNo: 105,
  name: 'Kiran',
```

```
> // Update one
db.students.updateOne(
  { rollNo: 101 },
  { $set: { marks: 92 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> // Update one
db.students.updateOne(
  { rollNo: 101 },
  { $set: { marks: 92 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
> db.students.deleteOne({ rollNo: 103 })
< {
  acknowledged: true,
  deletedCount: 1
}
> db.students.deleteOne({ rollNo: 103 })
< {
  acknowledged: true,
  deletedCount: 0
}
```

Welcome mongosh: 2520090003... +

> _MONGOSH

> db.students.find().sort({ marks: 1 })

< {

_id: ObjectId('6a8c01d38e356ef8a07b8765')

}

{

_id: ObjectId('6a8c015a8e356ef8a07b8761'),

rollNo: 107,

name: 'Ravi',

age: 23,

department: 'ECE',

marks: 68,

city: 'Nellore'

}

{

_id: ObjectId('6a8c01ec8e356ef8a07b8768'),

rollNo: 113,

name: 'Ramesh',

age: 22,

department: 'ECE',

marks: 79

}

{

_id: ObjectId('6a8c015a8e356ef8a07b8760'),

rollNo: 106,

name: 'Anu',

age: 20,

department: 'IT',

marks: 82,

city: 'Guntur'

}

{

_id: ObjectId('6a8c01ec8e356ef8a07b8767'),

rollNo: 112,

name: 'Pooja',

age: 20

> _MONGOSH

> db.students.find().sort({ marks: -1 })

```
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8764'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 92,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8762'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
```

> _MONGOSH

```
> db.students.find().limit(5)
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 92,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
  department: 'CSE',
  marks: 95
}
```

```
> db.students.find().sort({ marks: -1 }).limit(3)
```

```
< {  
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),  
  rollNo: 105,  
  name: 'Kiran',  
  age: 21,  
  department: 'CSE',  
  marks: 95,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a8c015a8e356ef8a07b8764'),  
  rollNo: 110,  
  name: 'Meena',  
  age: 20,  
  department: 'AI&DS',  
  marks: 93,  
  city: 'Hyderabad'  
}  
{  
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),  
  rollNo: 101,  
  name: 'John',  
  age: 20,  
  department: 'CSE',  
  marks: 92,  
  city: 'Hyderabad'  
}
```

```
> db.students.find().skip(5)
```

```
< {  
  _id: ObjectId('6a8c015a8e356ef8a07b8761'),  
  rollNo: 107,  
  name: 'Ravi',  
  age: 23,
```

>_MONGOSH

```
_id: ObjectId('6a8c015a8e356ef8a07b8762'),
rollNo: 108,
name: 'Sita',
age: 21,
department: 'CSE',
marks: 91,
city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8763'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8764'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS',
  marks: 93,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c01d38e356ef8a07b8765')
}
{
  _id: ObjectId('6a8c01e18e356ef8a07b8766'),
  rollNo: 111,
  name: 'Vijay',
  age: 21,
  department: 'CSE',
```

>_MONGOSH

> db.students.find().skip(5)

```
< {
  _id: ObjectId('6a8c015a8e356ef8a07b8761'),
  rollNo: 107,
  name: 'Ravi',
  age: 23,
  department: 'ECE',
  marks: 68,
  city: 'Nellore'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8762'),
  rollNo: 108,
  name: 'Sita',
  age: 21,
  department: 'CSE',
  marks: 91,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8763'),
  rollNo: 109,
  name: 'Arun',
  age: 22,
  department: 'AI&DS',
  marks: 87,
  city: 'Kurnool'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b8764'),
  rollNo: 110,
  name: 'Meena',
  age: 20,
  department: 'AI&DS'
```

>_MONGOSH

```
> db.students.find(
  {},
  { _id: 0, name: 1, department: 1, marks: 1 }
)
< {
  name: 'John',
  department: 'CSE',
  marks: 92
}
{
  name: 'Alice',
  department: 'CSE',
  marks: 90
}
{
  name: 'Priya',
  department: 'IT',
  marks: 88
}
{
  name: 'Kiran',
  department: 'CSE',
  marks: 95
}
{
  name: 'Anu',
  department: 'IT',
  marks: 82
}
{
  name: 'Ravi',
  department: 'ECE',
  marks: 68
}
{
```

>_MONGOSH

```
  name: 'Ravi',  
  department: 'ECE',  
  marks: 68  
}  
{  
  name: 'Sita',  
  department: 'CSE',  
  marks: 91  
}  
{  
  name: 'Arun',  
  department: 'AI&DS',  
  marks: 87  
}  
{  
  name: 'Meena',  
  department: 'AI&DS',  
  marks: 93  
}  
{}  
{  
  name: 'Vijay',  
  department: 'CSE',  
  marks: 86  
}  
{  
  name: 'Pooja',  
  department: 'IT',  
  marks: 84  
}  
{  
  name: 'Ramesh',  
  department: 'ECE',  
  marks: 79  
}
```

>_MONGOSH

```
> db.students.find(
  {},
  { age: 0 }
)
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  department: 'CSE',
  marks: 92,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),
  rollNo: 102,
  name: 'Alice',
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
  rollNo: 104,
  name: 'Priya',
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),
  rollNo: 105,
  name: 'Kiran',
  department: 'CSE',
  marks: 95,
  city: 'Hyderabad'
}
```

>_MONGOSH



```
    department: 'IT',
    marks: 82,
    city: 'Guntur'
  }
  {
    _id: ObjectId('6a8c015a8e356ef8a07b8761'),
    rollNo: 107,
    name: 'Ravi',
    department: 'ECE',
    marks: 68,
    city: 'Nellore'
  }
  {
    _id: ObjectId('6a8c015a8e356ef8a07b8762'),
    rollNo: 108,
    name: 'Sita',
    department: 'CSE',
    marks: 91,
    city: 'Hyderabad'
  }
  {
    _id: ObjectId('6a8c015a8e356ef8a07b8763'),
    rollNo: 109,
    name: 'Arun',
    department: 'AI&DS',
    marks: 87,
    city: 'Kurnool'
  }
  {
    _id: ObjectId('6a8c015a8e356ef8a07b8764'),
    rollNo: 110,
    name: 'Meena',
    department: 'AI&DS',
    marks: 93,
    city: 'Hyderabad'
  }
```

```
> db.students.aggregate([
  { $match: { marks: { $gt: 80 } } }
])
< {
  _id: ObjectId('6a8c015a8e356ef8a07b875b'),
  rollNo: 101,
  name: 'John',
  age: 20,
  department: 'CSE',
  marks: 92,
  city: 'Hyderabad'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875c'),
  rollNo: 102,
  name: 'Alice',
  age: 21,
  department: 'CSE',
  marks: 90,
  city: 'Vijayawada'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875e'),
  rollNo: 104,
  name: 'Priya',
  age: 22,
  department: 'IT',
  marks: 88,
  city: 'Warangal'
}
{
  _id: ObjectId('6a8c015a8e356ef8a07b875f'),
  rollNo: 105,
  name: 'Kiran',
  age: 21,
```

>_mongo>

```
> db.students.aggregate([
  {
    $group: {
      _id: "$department",
      totalStudents: { $sum: 1 }
    }
  }
])
< {
  _id: 'AI&DS',
  totalStudents: 2
}
{
  _id: 'ECE',
  totalStudents: 2
}
{
  _id: 'CSE',
  totalStudents: 5
}
{
  _id: 'IT',
  totalStudents: 3
}
{
  _id: null,
  totalStudents: 1
}
> db.students.aggregate([
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" }
    }
  }
])
```

```

> db.students.aggregate([
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" }
    }
  }
])
< {
  _id: 'AI&DS',
  averageMarks: 90
}
{
  _id: 'ECE',
  averageMarks: 73.5
}
{
  _id: 'CSE',
  averageMarks: 90.8
}
{
  _id: 'IT',
  averageMarks: 84.66666666666667
}
{
  _id: null,
  averageMarks: null
}

```

```

> db.students.aggregate([
  {
    $group: {
      _id: null,
      highestMarks: { $max: "$marks" }
    }
  }
])
< {
  _id: null,
  highestMarks: 95
}

```

```
> // Total marks
db.students.aggregate([
  {
    $group: {
      _id: null,
      totalMarks: { $sum: "$marks" }
    }
  }
])
< {
  _id: null,
  totalMarks: 1035
}
```

```

}
> db.students.aggregate([
  { $match: { department: "CSE" } },
  {
    $group: {
      _id: "$department",
      averageMarks: { $avg: "$marks" },
      totalStudents: { $sum: 1 }
    }
  },
  { $sort: { averageMarks: -1 } }
])
< {
  _id: 'CSE',
  averageMarks: 90.8,
  totalStudents: 5
}
```

```
> db.students.aggregate([
  { $sort: { marks: -1 } },
  { $limit: 3 },
  {
    $project: {
      _id: 0,
      rollNo: 1,
      name: 1,
      marks: 1
    }
  }
])
< {
  rollNo: 105,
  name: 'Kiran',
  marks: 95
}
{
  rollNo: 110,
  name: 'Meena',
  marks: 93
}
{
  rollNo: 101,
  name: 'John',
  marks: 92
}
studentDB>
```